Kyle Onghai

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Education

Princeton University — **Princeton, NJ**
Ph.D. Operations Research & Financial Engineering — *August 2024 – Present*
- Focus on stochastic control and mathematical finance and their applications to energy markets and decentralized finance.
- Advised by Professor Ronnie Sircar.

University of California, Los Angeles — **Los Angeles, CA**
B.S. Mathematics, Specialization in Computing — *September 2020 – June 2024*
- Highest Honors, Honors College, Alumni Scholars Club
- Top 500 on the 81st William Lowell Putnam Mathematical Competition

Research Experience

Princeton University — **Princeton, NJ**
Ph.D. Student — *May 2025 – Present*
- Forecasting data center electricity load growth and its effect on wholesale electricity prices through continuous-time stochastic processes.
- Analyzing a capacity expansion problem for a profit-maximizing central grid planner meeting uncertain demand using stochastic optimal control.
- Formalizing competitive, recursive subsidies powered by decentralized physical infrastructure networks (DePINs) for solar generators.

Massachusetts Institute of Technology — **Cambridge, MA**
Summer Geometry Initiative (SGI) 2023 Fellow — *July 2023 – August 2023*
- Modeled clogging within three-dimensional stochastic Poisson-Voronoi foams by calculating the distribution of edge lengths and simulating in Voro++.
- Generalized a method that generates helix-free stripes in knit graph designs to handle more than two boundaries and compute the relevant harmonic interpolations in Geometry Central.
- Devised a mesh decimation algorithm based on iterated local operations that adheres to preferred edge alignment directions in libigl.

Centre de Recherches Mathématiques — **Montreal, QC**
12th Montreal Industrial Problem Solving Workshop — *August 2022*
- Built a prototype for the International Air Transport Association to map air turbulence using real-time eddy dissipation rate data.
- Incorporated pilot feedback to optimize plots for interpretability.

Moravian University — **Bethlehem, PA**
REU: Research Challenges of Computational Methods in Discrete Mathematics — *June 2022 – August 2022*
- Introduced distribution graphs in the study of chip-firing games on simple graphs.
- Classified the contractions of non-supereulerian, 2-edge-connected graphs with matching number at most 4.

Collaborative Mathematical Research Program — **Online**
Undergraduate Researcher — *June 2021 – August 2021*
- Showed that, given $m \geq k$, there exists at least one k-Diophantine m-tuple over a finite field with a sufficiently large, odd prime cardinality.
- Proved a formula for the number of 3-Diophantine triples in a finite field with an odd prime cardinality.
- Developed an asymptotic formula for the number of k-Diophantine m-tuples in a finite field with an odd prime cardinality.

Stony Brook University — **Stony Brook, NY**
Simons Summer Research Fellow — *June 2019 – August 2019*
- Added photoacoustic tomography functionality to the Orthopaedic Bioengineering Lab's proprietary quantitative ultrasound machine.
- Adapted a numerical algorithm to reconstruct images of phantoms using Fourier transforms in MATLAB to the lab-specific hardware.

Publications and Presentations

Journal Articles

Hammonds, Trajan, Seoyoung Kim, Steven J. Miller, Arjun Nigam, **KO**, Dishant Saikia, and Lalit M. Sharma. "k-Diophantine m-tuples in finite fields". In: *International Journal of Number Theory* 19.04 (2023), pp. 891–912. DOI: 10.1142/S1793042123500458. arXiv: 2201.06232 [math.NT]. URL: https://doi.org/10.1142/S1793042123500458.

Preprints

Crosier, Alexander, **KO**, and Ronnie Sircar. *AI for AI: Optimizing Additional Infrastructure Build-out to Power Artificial Intelligence Data Centers*. 2026. arXiv: 2609.08166 [q-fin.GN]. URL: https://arxiv.org/abs/2609.08166.

Presentations

Fiorini, Eugene, Max Folger, **KO**, Jacob Porter, Danae Rupp, and Andrew Woldar. *Properties of Chip-Firing Distribution Graphs. Poster presentation at JMM 2023.*

Fornaci, Alberto, Adrian Ganea, Brent King, Katsiaryna Vashchankova, Javier Almonacid, Sean Bohun, Douglas Bowen, Thomas Gkelsinis, Slim Ibrahim, Michael R. Lindstrom, Joy Liu, and **KO**. *Creating a Heat Map and Building a Seasonal Diagram. Oral presentation at the 12th Montreal Industrial Problem Solving Workshop Concluding Symposium.*

Hammonds, Trajan, Seoyoung Kim, Steven J. Miller, Arjun Nigam, **KO**, Dishant Saikia, and Lalit M. Sharma. *k-Diophantine m-tuples in finite fields. Recorded oral presentation at the Young Mathematician Conference (August 2021) at The Ohio State University.*

Keitly, Benjamin, Samuel Murray, **KO**, Tony W. H. Wong, and Taoye Zhang. *Non-Supereulerian, 2-Edge-Connected Graphs with Matching Number at Most 4. Poster presentation at JMM 2023.*

Online Articles

Adapala, Sanjana, Sara Ansari, Shalom Abebaw Bekele, **KO**, and Andrew Spielberg. *Probabilistic Design and Analysis of Lattice Structures*. Available at https://summergeometry.org/sgi2023/probabilistic-design-and-analysis-of-lattice-structures/. Oct. 2023.

Altintas, Gizem, Emi Neuwalder, **KO**, Hossam Saeed, and Frank Yang. *Simplifying Meshes Along Frame Fields.*

Kargi, Sahana, **KO**, Anthony Ramos, and Edward Chien. *Knitting Foliations.*

Teaching and Community Involvement

Princeton University — **Princeton, NJ**

Teaching Assistant — *August 2025 – Present*

- Teaching Probability and Stochastic Systems (ORF 309) through weekly precepts and office hours.

Massachusetts Institute of Technology — **Cambridge, MA**

Summer Geometry Initiative (SGI) 2024 Volunteer — *July 2024 – August 2024*

- Facilitated research dialogue between mentors and fellows on various topics within geometry processing, including medical imaging and architectural algorithms.

Skills

Software: C/C++, Python, LaTeX, MATLAB, Lua, GNU/Linux

Languages: English (Native), Mandarin (Intermediate), Latin